Classical Field Theory — Exercise Sheet 7

Modes, complex pairs and admissibility

Solutions

Use $\{F,G\} = \int (F_\phi G_\pi - F_\pi G_\phi)$ and $\Omega = \int \delta\phi \wedge \delta\pi$. Except for the explicitly finite torus model in Problem 1, functionals have Lecture 7's smooth Schwartz-valued gradients. A formal distributional derivative does not automatically belong to that class.

Exercise 7.1 (Real oscillators in a finite torus model). On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, fix an integer $N \geq 2$ and use the real basis

$$e_0 = (2\pi)^{-1/2}, \qquad e_{n,c} = \pi^{-1/2}\cos(nx), \qquad e_{n,s} = \pi^{-1/2}\sin(nx), \quad 1 \leq n \leq N.$$

Let $\phi = q_0 e_0 + \sum_{n=1}^{N}(q_{n,c}e_{n,c} + q_{n,s}e_{n,s})$ and let π have the same expansion with coefficients p.

(a) Verify orthonormality and compute the pulled-back canonical form and the Hamiltonian $H = \frac{1}{2}\int_{\mathbb{T}}(\pi^2 + \phi_x^2 + m^2\phi^2)\mathrm{d}x$, where $m \geq 0$. Derive the oscillator equations and treat the zero mode at $m = 0$.

(b) Define $\widehat{\phi}_n = (2\pi)^{-1/2}\int e^{-inx}\phi\mathrm{d}x$, and similarly $\widehat{\pi}_n$, for $|n| \leq N$. Express them in the real coordinates and compute their complex bilinear brackets, including $n = 0$.

(c) Compute all nonzero real coefficients and the energy of the data $\phi = 3\cos(2x) - 2\sin x$, $\pi = 5\sin(2x) + \cos x$.

Solution. The elementary integrals of $\cos(nx)\cos(lx)$ and $\sin(nx)\sin(lx)$ are $\pi\delta_{nl}$ for positive indices, while every sine-cosine integral and every nonconstant mean vanish. The constant has squared integral one with the stated normalization. Substituting both expansions into the form therefore gives

$$\Omega_N = \delta q_0 \wedge \delta p_0 + \sum_{n=1}^{N}\sum_{\alpha=c,s}\delta q_{n,\alpha} \wedge \delta p_{n,\alpha}.$$

Spatial differentiation interchanges the sine/cosine directions with coefficients n and $-n$. Thus

$$H_N = \frac{1}{2}(p_0^2 + m^2 q_0^2) + \frac{1}{2}\sum_{n=1}^{N}\sum_{\alpha=c,s}\left[p_{n,\alpha}^2 + (n^2+m^2)q_{n,\alpha}^2\right].$$

Hamilton's equations are $\dot{q} = p$, $\dot{p} = -(n^2+m^2)q$ for each nonzero harmonic, and $\dot{q}_0 = p_0$, $\dot{p}_0 = -m^2 q_0$. At $m = 0$ the constant mode is a free particle: $q_0(t) = q_0(0) + tp_0(0)$, $p_0(t) = p_0(0)$.

For positive n, direct integration gives

$$\widehat{\phi}_n = \frac{q_{n,c} - iq_{n,s}}{\sqrt{2}}, \qquad \widehat{\phi}_{-n} = \frac{q_{n,c} + iq_{n,s}}{\sqrt{2}}, \qquad \widehat{\phi}_0 = q_0,$$

and the same formulas for $\widehat{\pi}$ in terms of p. The real canonical bracket has $\{q_{n,\alpha}, p_{l,\beta}\} = \delta_{nl}\delta_{\alpha\beta}$. Complex bilinear extension gives zero for two equal positive modes, since $1 + (-i)^2 = 0$, and one for opposite modes, since $1 + (-i)i = 2$. Distinct absolute indices give zero, and the constant bracket is one. Hence

$$\{\widehat{\phi}_n, \widehat{\pi}_l\} = \delta_{n+l,0}, \qquad \{\widehat{\phi}_n, \widehat{\phi}_l\} = \{\widehat{\pi}_n, \widehat{\pi}_l\} = 0.$$

No conjugation is inserted into this bilinear bracket.

For the last part the nonzero coefficients are $q_{2,c} = 3\sqrt{\pi}$, $q_{1,s} = -2\sqrt{\pi}$, $p_{2,s} = 5\sqrt{\pi}$ and $p_{1,c} = \sqrt{\pi}$. The kinetic contribution is $26\pi/2$, and the displacement contribution is $[9(4+m^2)+4(1+m^2)]\pi/2$. Their sum is $\pi(33+13m^2/2)$.

Exercise 7.2 (A complex scalar is two real canonical pairs). Write $\Phi = \phi_1 + i\phi_2$, with both real fields Schwartz, and take

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\nabla\Phi|^2 - m^2|\Phi|^2\big), \qquad m \geq 0.$$

Derive the two momenta and Hamiltonian by real variation. Compute all functional derivatives of $Q = \int(\phi_1\pi_2 - \phi_2\pi_1)\mathrm{d}^d x$ and its action on both fields and both momenta. Identify its finite flow, prove $\{Q, H\} = 0$, and verify admissibility in the product of the two real Schwartz phase spaces. Evaluate Q on $(\phi_1, \phi_2; \pi_1, \pi_2) = (f, 2f; -f, 3f)$, where $f \in \mathcal{S}(\mathbb{R}^d; \mathbb{R})$ and $\|f\|_2 = 1$.

Solution. The density is the sum of two real Klein–Gordon densities, each with factor $1/2$. Thus the Legendre maps give $\pi_a = \partial_t\phi_a$, and subtraction gives

$$H = \frac{1}{2}\sum_{a=1}^{2}\int_{\mathbb{R}^d}(\pi_a^2 + |\nabla\phi_a|^2 + m^2\phi_a^2)\mathrm{d}^d x.$$

For the charge, variation gives the four derivatives

$$Q_{\phi_1} = \pi_2, \quad Q_{\phi_2} = -\pi_1, \quad Q_{\pi_1} = -\phi_2, \quad Q_{\pi_2} = \phi_1.$$

Taking brackets in the order $\delta F = \{F, Q\}$ therefore gives

$$\delta\phi_1 = -\phi_2, \quad \delta\phi_2 = \phi_1, \qquad \delta\pi_1 = -\pi_2, \quad \delta\pi_2 = \pi_1.$$

Solving these two rotation systems gives simultaneous multiplication by $e^{i\alpha}$ of the complex field and complex momentum. The charge derivatives are continuous linear Schwartz maps. The Hamiltonian derivatives are $(-\Delta\phi_a + m^2\phi_a, \pi_a)$ as in Lecture 7. Both functionals are admissible in this product class.

For conservation, substitute the derivatives into the bracket to obtain

$$\{Q, H\} = \int\big[\pi_2\pi_1 - \pi_1\pi_2 + \phi_2(-\Delta\phi_1 + m^2\phi_1) - \phi_1(-\Delta\phi_2 + m^2\phi_2)\big]\mathrm{d}^d x = 0.$$

The kinetic and mass terms cancel directly. Integrating each Laplacian term by parts leaves opposite copies of $\nabla\phi_1 \cdot \nabla\phi_2$; Schwartz decay removes the boundary terms. For the supplied data, the charge density is $3f^2 + 2f^2 = 5f^2$, so $Q = 5$. The real normalization $\Phi = \phi_1 + i\phi_2$ was used throughout; no additional $1/\sqrt{2}$ or Wirtinger derivative is needed.

Exercise 7.3 (A smooth window and a sharp interval). In one dimension let $w \in C_c^\infty(\mathbb{R}; \mathbb{R})$ be a fixed window and set

$$F_w[\phi, \pi] = \frac{1}{2}\int_{\mathbb{R}} w(x)\phi_x(x)^2\mathrm{d}x, \qquad F_\#[\phi, \pi] = \frac{1}{2}\int_0^1 \phi_x(x)^2\mathrm{d}x.$$

Compute both derivatives against arbitrary real Schwartz variations. Prove F_w is admissible. Is $F_\#$ an admissible functional on the whole Schwartz phase space? Test the field $\phi(x) = (x + x^2)\eta(x)$, where $\eta \in C_c^\infty(\mathbb{R})$ equals one on a neighbourhood of $[0, 1]$. Display its endpoint distribution terms.

Solution. For the smooth window, differentiation and integration by parts give

$$DF_w[\phi,\pi](h_\phi,h_\pi)=\int_{\mathbb{R}} w\phi_x\partial_x h_\phi \mathrm{d}x=-\int_{\mathbb{R}}\partial_x(w\phi_x)h_\phi \mathrm{d}x.$$

Compact support of w removes the boundary term. Multiplication by a fixed smooth compact function and spatial differentiation are continuous on Schwartz space. Thus the displayed gradient is a smooth linear Schwartz-valued map, proving admissibility.

For the sharp functional the same step instead gives

$$DF_\#[\phi,\pi]h=\phi_x(1)h_\phi(1)-\phi_x(0)h_\phi(0)-\int_0^1\phi_{xx}h_\phi \mathrm{d}x.$$

Consequently its distributional field derivative is $-\mathbf{1}_{(0,1)}\phi_{xx}+\phi_x(1)\delta_1-\phi_x(0)\delta_0$; its momentum derivative is zero. At the chosen field, $\phi_x(0)=1$, $\phi_x(1)=3$ and $\phi_{xx}=2$ on the interval. Take smooth test bumps of height one at 1, supported in intervals of radius ε around 1, with uniformly bounded magnitude and zero value at 0. The derivative tends to 3 as $\varepsilon\downarrow 0$. Pairing any fixed smooth candidate gradient with these bumps tends to zero because their supports have length 2ε. This contradiction excludes a Schwartz gradient there. The quadratic functional itself is smooth on Schwartz space, but smoothness alone is weaker than Lecture 7's admissibility requirement.

Exercise 7.4 (The sign of the Hamiltonian commutator). For admissible F,G, let $H_F=D(\operatorname{grad}F)$ and $\mathbb{J}(a,b)=(b,-a)$ as in Lecture 7. Define the vector-field bracket by

$$[X,Y](z)=DY(z)X(z)-DX(z)Y(z),$$

the sign for which it acts on functionals as $X(YK)-Y(XK)$. Use the actual bracket-gradient formula of Lecture 7 to prove $X_{\{F,G\}}=-[X_F,X_G]$. For a nonzero example, choose real $f\in\mathcal{S}$ with $\|f\|_2=1$, put $q=\int f\phi$, $p=\int f\pi$, and take $F=q^2/2$, $G=p^2/2$. Compute both vector fields, their commutator, and $X_{\{F,G\}}$.

Solution. Write $a=\operatorname{grad}F$ and $b=\operatorname{grad}G$. Lecture 7's closure formula, proved using the symmetric Hessians, is

$$\operatorname{grad}\{F,G\}=H_F\mathbb{J}b-H_G\mathbb{J}a.$$

Since $\mathbb{J}$ is constant, $DX_F=\mathbb{J}H_F$ and $DX_G=\mathbb{J}H_G$. Therefore

$$[X_F,X_G]=\mathbb{J}H_G\mathbb{J}a-\mathbb{J}H_F\mathbb{J}b=-\mathbb{J}\operatorname{grad}\{F,G\}=-X_{\{F,G\}}.$$

Every direction and every output remains Schwartz by admissibility; no extension to arbitrary smooth functionals is being assumed.

For the example, the normalization gives $\{q,p\}=\int f^2=1$. The gradients and vectors are

$$\operatorname{grad}F=(qf,0),\quad X_F=(0,-qf),\qquad \operatorname{grad}G=(0,pf),\quad X_G=(pf,0).$$

The bracket is $\{F,G\}=qp$, whose vector is $(qf,-pf)$. Differentiating the displayed vectors gives $DX_G[X_F]=(-qf,0)$ and $DX_F[X_G]=(0,-pf)$. Their difference is $(-qf,pf)$, as claimed. At the allowed datum $(\phi,\pi)=(f,2f)$ it is $(-f,2f)\neq 0$, so this example detects an incorrect plus sign. The sign agrees with $X_FK=\{K,F\}$.